

Laboratory Protocol:

DNA Extraction from Plant Cells

Educational experiment for high school students

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1. Introduction

This protocol describes a safe and relatively simple procedure for extracting DNA from plant cells using easily available materials. The experiment is designed for high school students and takes approximately 45–60 minutes to complete.

2. Learning Objectives

- Understand the cellular structure of plant cells
 - Learn the basic principles of DNA extraction
 - Develop practical laboratory skills
 - Apply the scientific method
 - Understand the importance of DNA in modern biology
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3. Required Materials

For each group (3–4 students):

Biological material:

- 1 ripe banana (or other fruits such as kiwi or strawberries)
- Distilled water (100 ml)

Reagents:

- Liquid dish detergent (15 ml)
- Table salt (1 teaspoon)
- Cold 95% ethanol (must be kept in the freezer for at least 30 minutes)

Equipment:

- 1 mortar and pestle
- 2 × 250 ml beakers
- 1 × 100 ml graduated cylinder
- 1 coffee filter or gauze
- 1 funnel
- 1 test tube
- Glass stirring rod
- Plastic wrap
- Latex gloves
- Safety goggles
- Thermometer

4. Safety Measures

Before starting the experiment, it is essential to:

1. Wear protective gloves
2. Use safety goggles
3. Tie back long hair
4. Do not eat or drink in the laboratory
5. Wash hands before and after the experiment
6. Keep the workspace clean and organized

5. Detailed Procedure

Phase 1: Preparation of plant material

1. Peel the banana and cut about 3 cm
2. Place the banana piece in the mortar
3. Add 10 ml of distilled water
4. Gently crush with the pestle for 3–5 minutes until a homogeneous puree is obtained

Phase 2: Preparation of the extraction solution

1. In a beaker, prepare the extraction solution:
 - 100 ml distilled water
 - 15 ml dish detergent
 - 1 teaspoon of salt
2. Mix gently to avoid foam formation
3. Ensure the temperature is between 20–25°C

Phase 3: Extraction

1. Add 30 ml of the extraction solution to the banana puree
2. Gently mix with a glass rod for 5 minutes
3. Filter the mixture through a coffee filter placed in a funnel
4. Collect the filtrate in a clean beaker

Phase 4: DNA precipitation

1. Carefully transfer the filtrate into a test tube (about 1/3 full)
2. Slightly tilt the test tube
3. Slowly pour cold ethanol along the wall (same volume as the filtrate)
4. Observe the formation of an interface between the two phases

6. Observations and Expected Results

- A whitish layer will form at the interface between the aqueous solution and the alcohol
 - The precipitated DNA will appear as white floating filaments
 - The DNA can be collected using a glass rod
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7. Scientific Explanation

The extraction process is based on several principles:

1. Mechanical disruption of cells:

- Grinding breaks the cell walls
- Releases cellular contents, including DNA

2. Action of detergent:

- Breaks cell membranes
- Dissolves nuclear membranes
- Denatures proteins associated with DNA

3. Role of salt:

- Neutralizes the negative charges of DNA
- Facilitates precipitation

4. Function of alcohol:

- DNA is insoluble in alcohol
 - Causes DNA precipitation
 - Separates DNA from other molecules
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8. Further Questions

1. Why is it important that the alcohol is cold?
 2. What role does the detergent play in the extraction?
 3. Why is salt used in the extraction solution?
 4. What would happen if the solution were not filtered?
 5. Why is it possible to see DNA with the naked eye in this experiment?
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9. Troubleshooting

Problem: DNA is not visible

Possible causes:

- Alcohol not cold enough
- Mixture too agitated
- Incorrect reagent concentrations

Problem: DNA appears dark or contaminated

Possible causes:

- Inadequate filtration
 - Excessive mixing
 - Sample degradation
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10. Appendix**Protocol variations**

Other plant materials can be used:

- Kiwi
- Strawberries
- Onion
- Peas

Suggestions for the teacher

1. Prepare cold alcohol in advance
 2. Divide the class into groups of 3–4 students
 3. Allocate 15 minutes for the initial discussion
 4. Dedicate 30–40 minutes to the experiment
 5. Reserve 15 minutes for the final discussion
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Assessment

Suggested evaluation criteria:

1. Active participation
2. Accuracy in execution

3. Understanding of principles
 4. Quality of observations
 5. Ability to answer questions
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Bibliography and additional resources

1. Campbell, N. A., & Reece, J. B., *Biology* (latest edition)
2. Alberts, B. et al., *Molecular Biology of the Cell* (latest edition)
3. Online resources from EMBL (European Molecular Biology Laboratory)